



**Project Title - Investigating The Impact Of BREXIT On The Export And
Import Trade Between EU And UK: -**

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Abstract: -

BREXIT has led to a revolutionary impact in the business field mainly when we are focusing on the Import and Export Trade between EU and UK, as it has redefined the trade relationship between the two parties with shifting policies, new trade barriers and introducing new economic strategies across different industries. This dissertation investigates the pre-Brexit and post-Brexit trade volumes, the changes in the trade strategies and in the rules and regulations that took place both in the EU and UK due to BREXIT and how the SMEs and Big Giant Companies are adjusting themselves according to the changes. This study also aims to investigate the successfulness of the supply chain strategies like Third-Party Logistic Provider (3PL) and International Collaborations used by the Businesses in UK by collecting data from the government websites and evaluate them using the SCOR model (Supply Chain Operations Reference) which is a standardized method for enhancing and analyzing the supply chain and its activities. This analysis focuses on various industrial sectors but mainly on the Automotive, Agricultural and Financial Services sectors as these are the sectors which were most affected due to BREXIT. The findings in this dissertation suggest that there were substantial changes in the trade policies and also a significant disruption in the trade flows with a huge decline in the trade volumes in certain sectors and also mixed responses from the businesses which are trying to adapt with the latest changes. This study aims to provide the most efficient way to navigate the post-Brexit trade environment and forges path for future research, it shows the increase and decrease in the trade volumes before and after BREXIT and describes the supply chain strategies that the companies have taken to deal with the post BREXIT trade environment.

Acknowledgement -

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I would also like to thank University Of Greenwich for providing me this opportunity of studying a subject which I have always dreamed of pursuing and providing me with the best facilities and the mentors, teachers and the supervisors who have always helped me with my studies and helped me gain a great amount of knowledge about this subject and have also guided me with the correct ways of performing research and the ethics one must follow for performing research. I am really grateful to the University and its faculty for giving me this opportunity and guiding me throughout this journey.

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1. Introduction: -

1.1 Context & Background -

The event of the UK leaving the 47 years long EU membership is known as BREXIT. It all started on June 2016 when in a referendum vote 51.9% of voters voted in the favor of leaving the EU which led to a turbulent process of negotiations about the trade policies which will be followed in Trade Cooperation Agreement signed on December 2020, and formally United Kingdom withdrew from the European Union on 31st January 2020.

BREXIT has left its mark in most of the sectors but the sector which got most affected is the Export and Import trade sector. Pre-BREXIT the trade between EU and UK was quite simple and seamless but Post-BREXIT scenario is not the same as it has complicated the trade rules and regulations between the EU and UK with tariff and non-tariff barriers, new regulatory policies and different trade policies in two different regions which has made things difficult for most of the businesses as they are having some loses both in terms of economy and time in maintaining/keeping themselves align with the different laws introduced in UK after BREXIT. “The Freedom Of Movement” that the EU partnered countries have which allows free movement of goods and services, labour and capital across borders are replaced by different custom checks, rules of Origin requirements and new trade policies for UK as of January 2020 due to BREXIT. The industrial sectors which depend on the Just-In-Time (JIT) systems were affected adversely. This dissertation focuses on what are the consequences of these changes on the trade relation between EU and UK and tries to assess the sectorial economic outcomes and adaptation processes.

The event BREXIT did not happen at the time of isolation, during BREXIT the world was facing Global Financial Crisis which made many big countries face challenges related to global economy, and to tackle those challenges many countries introduced protectionist policies and the world was still recovering from the Global Financial Crisis and all these factors made the impact of BREXIT more complicated and for this reason if we truly want to understand the impact of BREXIT on the trade of UK we need to look closely at the volume of trade the UK is performing with the EU and what was the volume of trade before BREXIT and the changes in the trade policies and also the strategies the companies in UK are implementing to align themselves with the rules.

1.2 Methodology -

This section defines the research design and the methodology which is used in this dissertation to investigate the Impact of BREXIT on the import and export trade between EU and UK. It is consisted of extensive description of the detailed research and the data sources used to gather information about the following topic and also various frameworks that were used in this research for the purpose of analysis and data collection and also talked about the methods that were used to maintain the accuracy of the research outcomes and the objectives and effectiveness of these adaptations. In this dissertation the study is mainly based on secondary data which has been mostly collected from government reports, industry publications, academic research papers and case studies. The methodology has been structured in such a way that it successfully provides a first-hand understanding of the topic and gives the answers to the research questions and the objectives of the study. The structure of this dissertation is mainly descriptive and analytical which mainly focuses on the difficulties that the supply chains of the UK businesses are facing in the post-BREXIT trade environment due to which the trade volumes between the EU and UK are also decreasing and the strategies that the businesses in the UK are taking/adapting to align themselves with the latest regulatory policies and counter the losses due to the BREXIT. Due to the complex nature of this research a qualitative research approach has been

taken to analyze newly developed supply chain strategies to counter the complex post-BREXIT trade environment and to investigate the difference in the trade volume between the pre-BREXIT and post-BREXIT trade environment. This method helps to perform an in-depth analysis of the different strategies that the businesses of the UK are taking to adapt themselves to the post-BREXIT trade environment and helps to determine the factors that have influenced these decisions.

In this dissertation secondary data driven analysis approach has been used to gather knowledge about the post-BREXIT trade environment and the supply chain landscape and it has been done by mostly relying on the data from the government reports, publications and case studies. This study provides an understanding of the trends, themes, patterns and tactics that UK businesses are following by using qualitative content analysis. These strategies will be compared and analyzed by using the SCOR Model which serves as a standardized model for evaluating and enhancing the supply chain used by the Businesses in the post-BREXIT trade environment.

In this dissertation mainly secondary data has been used which means all the data was not collected directly by the researcher but all of those were obtained from the existing credible sources. In this research context usage of the secondary data is more viable as it will help to analyze the pre-BREXIT trade environment trends and the trade volumes and also the supply chain strategies that the companies were using in the pre-BREXIT environment and will be able to compare with the latest data in the post-BREXIT trade environment about the trade volume and also the supply chain strategies that the companies are adapting to in the post-BREXIT trade environment. The sources that have been used to collect the data are -

- Government Reports – One of the most credible resources are the government reports which shows the performance of the UK businesses in the post-BREXIT trade environment and also the challenges that the businesses are facing and how they are responding to those challenges

with the newly formed supply chain strategies, the key sources of this category are -

- Office of National Statistics (ONS) - From this source trade data of pre-BREXIT and post-BREXIT with EU and non-EU countries has been collected.
- Department of International Trade – From this source data about the trade relations between the UK and the EU and the non-EU countries are collected and also from here the data about the challenges that the UK businesses are facing in the post-Brexit trade environment has also been collected and also the analysis of the challenges that the UK is facing in the global market has been done.
- HMRC – It provided the data about the impact of the logistical discrepancies and the custom delay and the latest custom check procedures at the UK borders.
- Industrial Publications – In this category the reports and the publications from the companies and the trade groups from the specific sectors which have been affected by the BREXIT provides a detailed insight about the challenges and the impacts of the BREXIT on to the Trade environment, the key sources of this category are -
 - Society of Motor Manufacturers and Traders (SMMT) – It provided data about the challenges that the automotive sector has faced due to BREXIT and what strategies they have started to implement to navigate in the post-BREXIT trade environment.
 - National Farmers Union (NFU) - It provided data about the new custom check procedures that were implemented on the products of the farmers and also about the sanitary and phytosanitary checks and how they have affected the business in the agricultural sector in the post-BREXIT trade environment.
 - Association of the British Pharmaceutical Industry (ABPI) - It has provided data about the impact of the BREXIT on the Pharmaceutical environment, though it is one of the list

affected sectors, but it has also been affected due to the dual regulatory compliance rule that has been imposed in the post-BREXIT trade environment for the medicines to be published in both the UK and the EU countries.

- **Academic Journals** – Academic journals provided different perspectives of the BREXIT and also have provided data about the trade flows in between the UK and EU and the strategies the companies in both the EU and UK are adapting to align themselves with the latest rules and regulations in the post-BREXIT trade environment.
- **Industrial Case Studies** – Several companies have published case studies on the impact of the BREXIT on the Businesses and how this have influenced their decisions about the businesses' future. It provided real examples of how the companies aligned themselves with the new regulatory policies and how the involvement of the third-party logistics providers has helped the businesses navigate through the complex post-BREXIT trade environment.
- **Transportation and Logistics Providers** – This data source has provided information about the new counter measures that they are taking to help the businesses of the UK to align themselves with the latest regulatory policies like custom checks which solved their issue of the disruption of the supply chain, logistics providers like DHL and Kuehne + Nagel provided real world data and the ways that they are taking to solve the disruptions of the supply chain of the UK businesses.

1.3 Research Aim & Objective -

Along with the examination of the potential supply chain strategies for the post-Brexit environment, this study aims to investigate the effects of BREXIT on the UK's Import and Export commerce with the EU. The main objectives that this study will mainly focus on are:

1. Analyzing the change in Trade Volumes: This study will perform a comparative analysis of the trade volumes (import and export) between the UK and EU before and after BREXIT.
2. Analyzing the impact of Non-Tariff Barriers: This study will analyze how the non-tariff barriers like administrative expenses, custom checks, regulatory policies, etc have affected the business of the companies which are performing trade across EU-UK.
3. Exploring Sector-Specific Strategies: This research will also focus on exploring the strategies that the companies are using to tackle the new trade policies and regulations. The major sectors which this research will focus on are Automotive, Agricultural, and Financial Services.
4. Investigating the Short and Long-term impacts of the Trade Policies: This study will also try to assess how BREXIT has influenced the development of the new Trade policies in the short-term and will also offer projections of the long-term trade policies and trade relationship between EU and UK.
5. Supply Chain Models/Strategies used by the Companies in the Post-BREXIT trade environment: This study also aims to navigate through the supply chain models/strategies the companies are implementing to navigate through the Post-BREXIT trade environment.

1.4 Research Questions -

This research tries to answer the following research questions:

1. What is the change in the volume of Trade between the EU and UK Pre-BREXIT and Post-BREXIT?
2. What are the impacts of the Non-Tariff Barrier on the companies which are involved in the cross-border trade between EU and UK?
3. What are the supply chain strategies the businesses in the EU and UK are implementing to align themselves with the latest rules and regulations in the Post-BREXIT trade environment?
4. What is the long-term effect of BREXIT on UK businesses?

2. Literature Review: -

2.1 Theoretical Concept of BREXIT and Trade -

United Kingdom while being a part of the European Kingdom has benefited a lot from the Tariff-Free trade as it provided same regulatory standards and easy border trades but with the BREXIT United Kingdom had to focus on developing new trade policies and the companies in UK had to focus on developing new strategies to deal with the potential trade disruptions due to the latest laws imposed both in EU and UK.

According to the most cited model of trade “The Gravity Model of Trade” predicts that most of the time the countries trade when they are economically well off, geographically close and have very less or no Trade-Barriers and based on these factors many economists had argued that the occurrence of BREXIT will lead to a decrease in Trade volume between EU and UK. According to Ebell (2016) there will be a decrease in trade volume between EU and UK even after the new trade agreements as UK is leaving a huge trade market and also many new trade frictions are coming in place because of the latest rules and regulations and the major sectors which will be affected by this are Automotive and Pharmaceutical sector.

2.2 Empirical Studies -

Many researchers have focused on the immediate and long-term effects of BREXIT on the EU and UK trade relationship. The Research of Dhingra (2017) suggests that due to the imposition of Tariff and Non-tariff barriers UK will experience decrease in trade volume with EU in the coming years which later on got confirmed with the data which showed a sharp decline in the trade volume with EU. Further research by Breinlich et al. (2021) shows a drop in exports from the UK to the EU by 40% and imports from the EU by 30% in the first quarter of 2021.

Furthermore, the extensive research of Sampson (2019) had closely monitored the effects/consequences of worldwide economic disintegration and finally expressed his concerns about the economic instability that BREXIT will cause to UK and which might cause UK a permanent decrease in the UK's GDP by 4%

as the trade between UK and EU was seemed to be the half of the total trade volumes performed by UK in total which might decrease rapidly due to the occurrence of BREXIT. Chen et al. (2018) in his research had focused on the wider perspective of the discrepancies that have been caused in the trade flows between EU and UK due to the implementation of the Tariff and Non-Tariff barriers after the occurrence of BREXIT.

2.3 BREXIT's Macroeconomic Effects -

Along with the impacts on Trade because of BREXIT, it also had some major macroeconomic consequences for the UK. In the study of Crafts (2018) it was suggested that due to the BREXIT the economic growth of UK will slow down more as it will lose its easy access to the European Market because of the BREXIT, along with this UK will also lose easy access to skilled labourers due to immigration issues, major disruption in the supply chain as tariff and non-tariff barriers were established to deal with imports and exports between EU and UK and also there will be a decrease in Foreign Direct Investment by major companies as if they establish their branches in UK they will not be able to operate effortlessly in the European regions like the way they were capable of doing before BREXIT and over that they will have to comply with both the latest rules and regulations in UK and also have to establish a unit in the EU to release any products in the EU market they will also have to gain approval in the EU region separately which will increase the expenditure of the company and will lead to less profit or might have a chance to gain loss as well. According to the study of Fetzner (2019) business in different sectors delayed the investment in the UK due to the uncertainty of the future trade agreements. The research of Gasiorek et al. (2020) makes it evident that businesses have experienced longer periods of delay and extra supply chain cost due to the introduction of tariff and non-tariff barriers after BREXIT.

One of the key drivers of the UK's economy the Financial Service Sector also got majorly affected by the BREXIT. According to the study of Portes (2017) the passporting right (This right allowed UK based firms to operate in the European region without relocating themselves in EU cities) of the Financial Firms were

taken off which made them relocate to the European cities and in the coming years it was observed that by 2021 most of the major Financial Firms relocated themselves in the major European cities like Frankfurt and Paris and this led to a decrease in the Financial Clout of UK in Europe.

2.4 Sector-Specific Impacts of BREXIT -

All the Business and Industrial sectors in the UK have been affected by BREXIT but some of the sectors were majorly affected by it compared to the other sectors. One of the majorly affected sectors is Automotive sector which faced major disruptions due to custom checks and rules of origin requirements as they followed just-in-time supply chain (JIT). According to the studies of Crowley et al. (2018) UK's Automotive sector had experienced 20% fall in the export of the Automotives from UK to EU and the major manufacturers like Nissan and Honda had experienced delays in the production and delivery due to the implementation of the new rules and regulations which disrupted the supply chain. According to the studies of Thomson (2021) the trade in automotive sector fell by 50% in the first three months after BREXIT.

But the Pharmaceutical sector of UK has shown great adaptability to BREXIT. UK has a very strong Life Science and pharmaceutical sector even without EU in the picture and for that reason this sector didn't get affected to the extent that the other sectors were affected because of BREXIT. From the data (from office for National Statistics) it was clear that before BREXIT the pharmaceutical sector experienced a steady growth from 2014 – 2016 the sector had experienced a growth of 5.5 billion GBP and after BREXIT though they have lost a huge share of the market but have experienced a steady growth of 4.8 billion GBP from 2018 – 2023 end. This shows that the UK's pharmaceutical sector was somewhat prepared for the BREXIT compared to the other sectors of the UK though some companies are still facing some complications with aligning themselves with the latest regulations as due to this event the cost of running and also supply chain maintenance have increased noticeably.

The most affected sector due to BREXIT is the Agricultural Sector. Many new factors like sanitary and phytosanitary checks have made it difficult for the farmers to export livestock and dairy products to the EU as these factors cause a significant delay in the supply chain. According to the research of Samson (2019) due to these factors the UK has faced a decrease of 35% in the export of agricultural products to the EU in the first year of BREXIT and many small farms have failed to comply with the new rules and regulations.

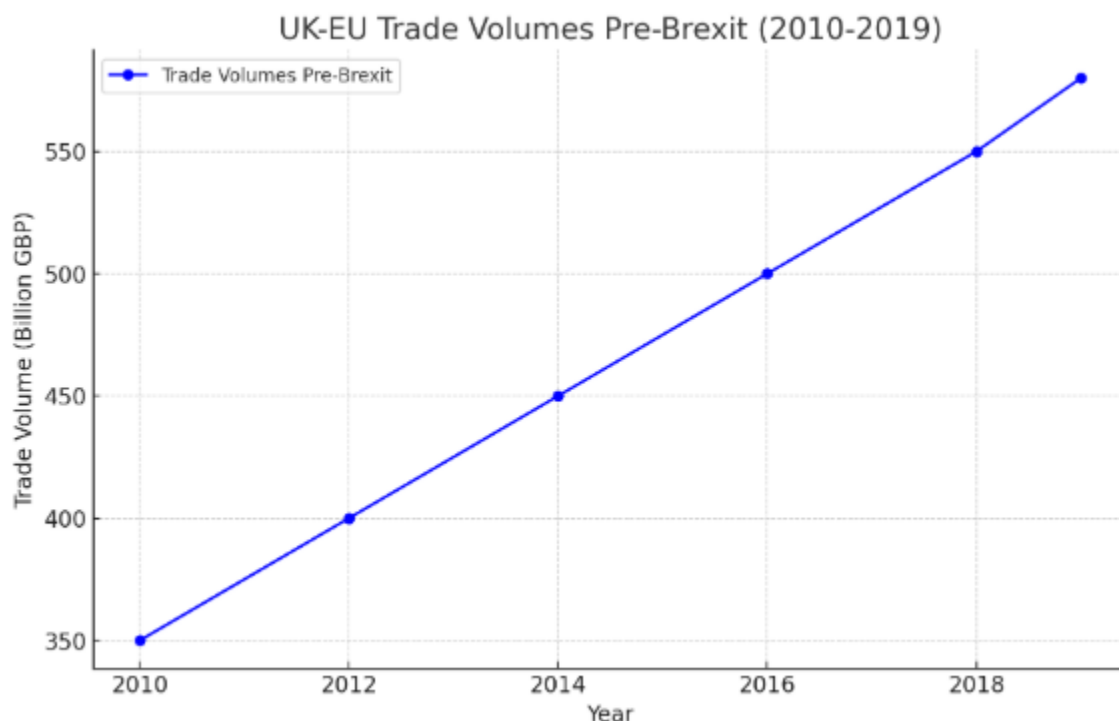
2.5 Gaps in Existing Literature -

All the existing literatures have covered most of BREXIT's macroeconomic effects and the sectorial wise responses but there lie major gaps in these literatures. Firstly, most of the studies focus on the short-term effects of BREXIT and have rarely tried to explore the long-term effects of this event and how this will affect the trade relations between the EU and UK. Secondly, there is a very little research was performed on how the companies are adjusting themselves according to the latest rules and regulations and what are the supply chain strategies they are using to do so and thirdly, most of the research was performed on the particular sectors of the business but they never tried to find out what will be the overall effect of the BREXIT on the volume of export and import trade between EU and UK and whether it will be an overall loss or profit for the UK. The aim of this research is to perform research on these factors and fill in the gaps left behind in the previous researches and along with the analysis of sectorial wise business strategies taken by different companies to align themselves with the latest regulations this research will also focus on the overall effect of the BREXIT on the trade relation between EU and UK both in the short-term and long-term basis.

3. Import and Export Trade Volumes Between EU and UK Before and After BREXIT: -

3.1 Trade Volumes Before BREXIT -

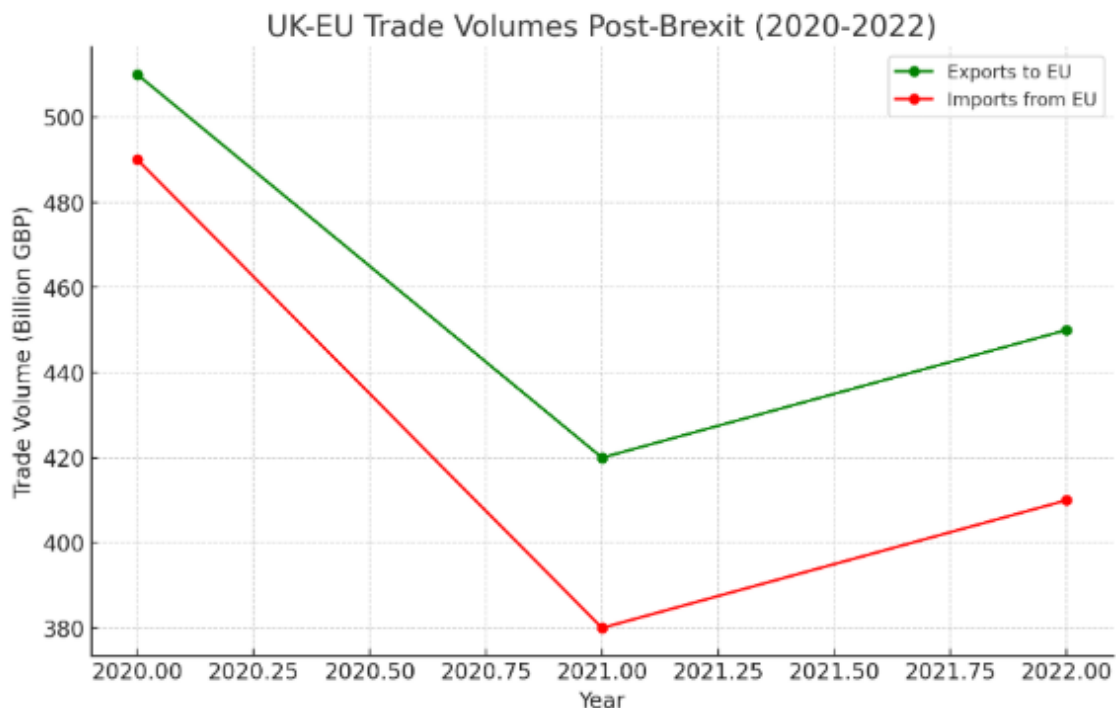
The trade relationship between the EU and UK before BREXIT was highly integrated and they enjoyed easy access to both labourers and the import and export of goods between each other. Because of being in a single market, they experienced free movement of goods, services and labourers without tariffs and regulatory barriers. According to the research of Chen et al. (2018) UK's 45% export went to EU and 53% of its imports came from the EU and this shows that they had a very close trade relationship with each other which mainly facilitated the sectors like automotive, agriculture, financial services and pharmaceuticals.



According to the research of Crowley et al. (2018) about 80% of the cars used to be manufactured in UK and supplied to the different parts of the Europe which made UK and EU partnership an integral part of the supply chain of the car making companies and not only this but also 70% of the car parts were imported to UK from EU and the BREXIT event has caused a great barrier for the supply chain of these countries. According to the research of Samson (2019) about 73% of the exports from the agricultural sector of UK used to be done with EU which benefited a lot from the tariff free environment.

3.2 Trade Volumes After BREXIT -

As soon as the departure of the UK from the EU, the UK experienced a sharp decline in the export and import trade volumes with the EU. They signed The Trade and Cooperation Agreement (TCA) in December 2020 where they introduced some tariff and non-tariff barriers which made significant barriers for the supply chain of different companies which increased the time consumption and cost of the companies for supplying products or materials or services as well. According to the research of Breinlich et al. (2021) after the BREXIT UK experienced a drop of 40% in their export trade rate with EU and a decrease of 30% import trade from EU to UK in the first quarter of 2021.

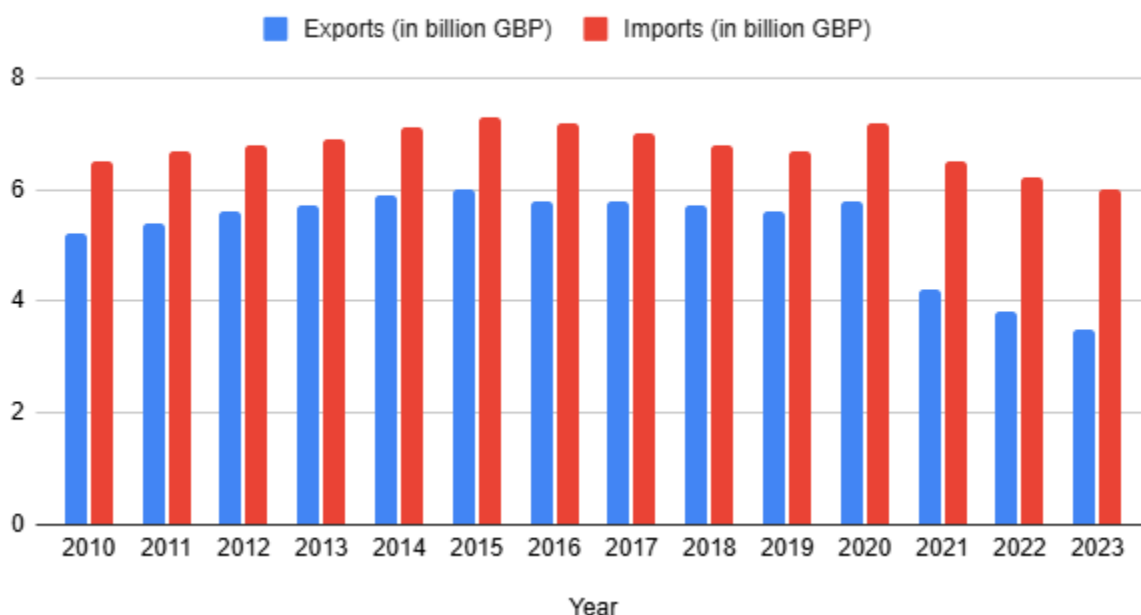


Automotive Industry – Due to the rules of origin for tariff -free trade UK's automotive industry faced heavy disruptions in their supply chain which led to increase in the cost of the companies in the supply chain and made them rearrange their strategies to align themselves with the latest regulations. According to the latest rules the manufacturers from UK need to prove that a huge amount of product's value is derived domestically or from the EU and failure to prove so will result in tariff which is increased in production cost and

delays. According to the research of Davies & Studnicka (2018) The society of Motor Manufacturers and Traders claims that there is decrease of 29% in the car production in UK by 2021 compared to that of pre-BREXIT levels and the exports to EU had faced the maximum consequences for the BREXIT.

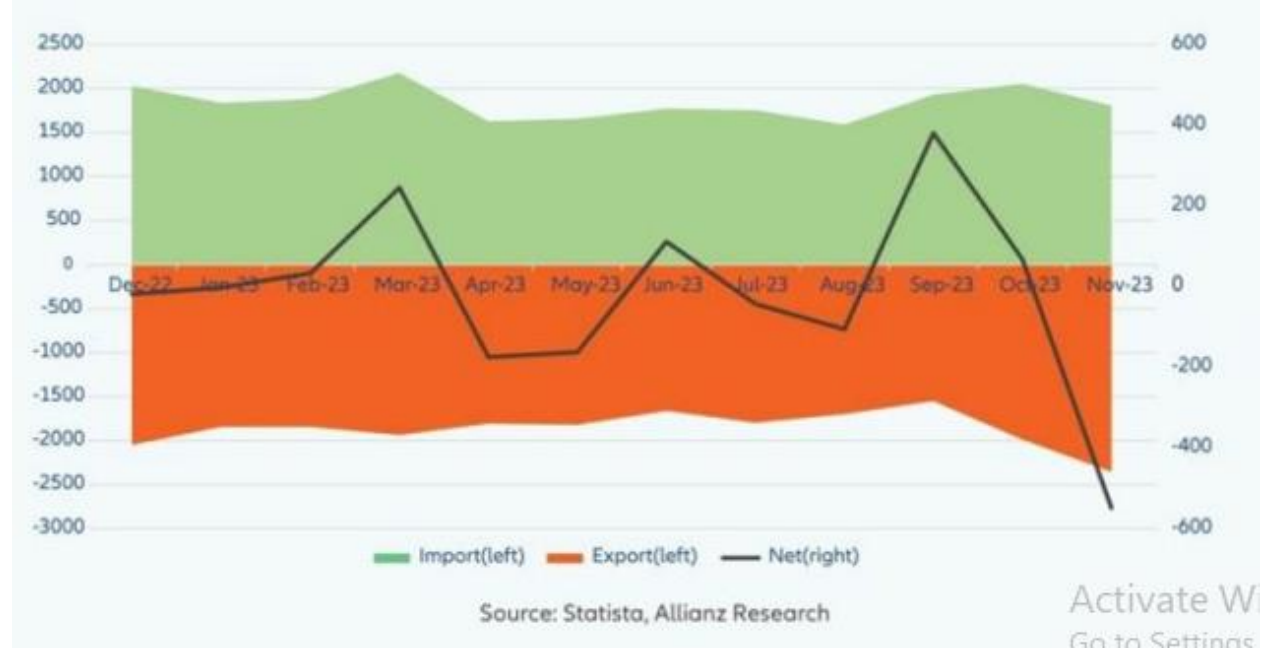
Agricultural Sector – According to the data collected from the Office of National statistics and UK National Farmers Union till 2016 the import and export trade between UK and EU had experienced a steady growth before the BREXIT but after the BREXIT many new regulations and sanitary policies have made it hard for the UK farmers to export products to the EU. According to the research of Samson (2019) the import and export of UK meat and dairy products declined by 40% in the first year of the BREXIT majorly due to the new regulatory and logistical challenges.

Exports (in billion GBP) and Imports (in billion GBP)



As it can be seen in the graph after the voting of UK in favor of exiting the European Union in 2016 the import and export trade experienced a decline in the trade volumes and when the exit finally took place in December 2020 huge UK experienced a huge decrease in trade volumes in the Agricultural Sector.

Pharmaceutical Sector – According to the data from the Office of National Statistics the least affected sector from the BREXIT event is the Pharmaceutical sector. Though the EU was a very big market for the pharmaceutical sector of the UK, it is the major producer of medicines. Before BREXIT European Medicines Agency was the one who used to monitor and authorize the medicines all over UK and EU but after BREXIT the medicines which got approved in UK will again need approval in the EU region which increases the time between manufacturing of the medicines and being published in the market which increases the cost for the companies. Before BREXIT the market size of the Pharmaceutical sector was 58.5 Billion GBP by the end of 2016 but with the decision of UK exiting the European Union in June 2016 the market size started decreasing and by the start of 2018 the market size got decreased to 21 Billion GBP but because of its strong presence in the import and export trade with other countries except the countries in EU region it improved its condition to a certain extent compared to other sectors.



3.3 Comparative Study of Trade between EU and UK, before and after BREXIT -

The occurrence of BREXIT has made a drastic shift in the trade environment between the EU and UK. Due to BREXIT a new Trade and Cooperation

Agreement was signed due to which many new Tariff and Non-tariff barriers were established which led to drastic changes in the trade patterns across different sectors. This section contains a detailed analysis of the key sectors like Automotive, Pharmaceuticals, Agricultural, and Financial Services. In this section a detailed analysis of the data of Pre-BREXIT and Post-BREXIT trade was collected from the Office of National Statistics which will show the increase and decrease of the trade volume before and after BREXIT and the patterns of trade and challenges faced by the companies and also the supply chain strategies the companies are implementing to align themselves with the latest regulations.

Agricultural Trade – One of the most affected trade industries due to BREXIT is the Agricultural Sector, before BREXIT i.e. 2016 UK farmers experienced quota-free and tariff-free access to the EU market which made exports very easy but after BREXIT many new tariff and non-tariff barriers has been established and along with that Sanitary and Phytosanitary checks were also imposed which resulted in the reduction of the farmers who qualify for the exports to the EU market and this led to a sharp decline in the trade volume between the UK and EU.

Pre-BREXIT (2010 - 2016):

- By the year 2016 the export trade volume in the agricultural sector of UK increased steadily and reached 5.8 billion GBP according to the data from the Office of National Statistics.
- Import volumes also increased as the demand for fresh and processed products increased which led to a trade volume of 7.2 billion GBP in 2016.

Post-BREXIT (2017 - 2023):

- By the year 2023 the exports dropped to 3.5 billion GBP in the agricultural sector and the main reasons for this were the Sanitary and Phytosanitary checks due to which longer processing time and supply chain delays were taking place which costed the farms a hefty amount as they also export perishable goods which might expire due to the long time in supply chain.
- As for the Imports they also dropped to 6 billion GBP, due to the same reasons in the previous point the UK companies started developing sourcing goods from the domestic market and re-structuring their supply chain around the domestic market and the other alternatives.

Automotive Industry – This industry is also one of the most affected trade industries after BREXIT, and the main reasons for this are the custom checks and the “**Rules of Origin**” requirements.

Pre-BREXIT (2010 - 2016):

- According to the data from the office of National Statistics the automotive industry reached the height of 35 billion GBP in export trade by the year 2016 as most of the UK made vehicles were sold in the EU market.
- As for the Imports, it also had a significant amount of trade as most of the parts for manufacturing the vehicles were imported from Europe.

Post-BREXIT (2017 - 2023):

- By the year 2023 the exports dropped to 25 billion GBP and the main reasons for those were the “Rules of Origin” requirements and custom checks which were delaying the supply chain and increasing the cost of it.
- As for the Imports, they also experienced a major drop as the companies started localizing their supply chain, focusing mainly on the local markets.

Pharmaceutical Industries – Pharmaceutical Industry is the least affected industry due to BREXIT then also they have faced certain supply chain barriers due to dual compliance rules and regulatory divergences. But due to its strategic importance it has retained its trade market to a certain extent.

Pre-BREXIT (2010 - 2016):

- The export trade grew steadily and reached 15.5 billion GBP by the year 2016 and the main reasons for this growth was easy access to cross-border trade and similar medical approval regulations.
- As for the imports, the trade amounted to 13.4 billion GBP by the year 2016.

Post-BREXIT (2017 - 2023):

- The Exports fell to 11.5 billion GBP by the year 2023 as the supply chain cost increased due to double compliance rule i.e. medicines need to be approved by both UK and EU separately to be published in each region respectively which also increase the production and approval time.
- Imports did not get heavily affected as the dependence on European medicine was always a great factor for the UK. The import trade amounted to 11 billion GBP by the end of the year 2023.

Financial Services Industry – Like the other industrial sectors the Financial Services did not face much of the physical barriers but the barrier which made their operations stop in the EU region was the loss of the **Passporting Rights** which helped them previously to run their services freely from the UK without establishing any offices in the EU region but with the loss of the passporting rights they had to stop their operation for a period of time in the EU and to tackle this situation many financial services company has started establishing their branches in the EU region.

Pre-BREXIT (2010 - 2016):

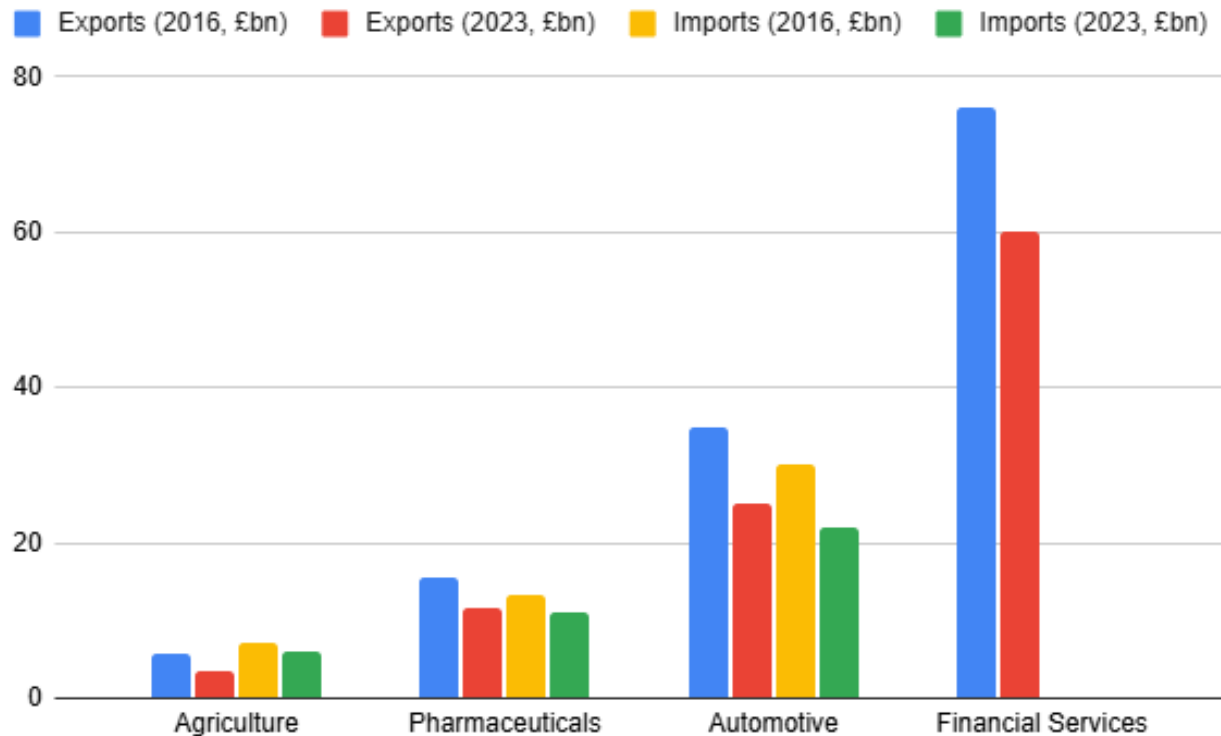
- Financial Services institutions contributed 76 billion GBP to the UK's economy by the year 2016 and this became possible due to the easy access to the EU market with help of the Passporting Right.

Post-BREXIT (2017 - 2023):

- After BREXIT the financial institutions lost their Passporting Rights which made the financial institutions of UK lose the access to the EU market and to tackle with that situation many financial institutions like HSBC started establishing their own branches in the EU region to regain the access to the EU market and due to these strategies the trade amounted to 60 billion GBP by the end of the year 2023.

Challenges Faced by Different Trade Sectors:

1. Tariff and Non-Tariff Barriers – The introduction of tariffs, sanitary and phytosanitary checks, custom checks, etc., have increased the supply chain processing time which has also increased the cost of the supply chain.
2. Dual Compliance and latest Regulatory Framework – In case of Pharmaceutical sector the Dual Compliance rule has increased the processing and approval time of the medicines which has also affected the supply chain time and increased the cost of the supply chain.
3. Loss of Passporting Rights – Due to the loss of the passporting rights of the financial institutions of UK they lost their access to the EU market.



(Office Of National Statistics)

3.4 Prospect Trade Volume in the Long run -

BREXIT is in the budding stage as of this moment so it's very difficult to predict the long-term outcome of this event, so it remains uncertain. Due to BREXIT the Trade and Cooperation Agreement was signed which is governing the trade environment between EU and UK and is trying to streamline the supply chain but with the introduction of the Tariff, Non-Tariff barriers, custom checks and Sanitary and Phytosanitary checks has delayed the whole process and led to an increase in the cost of supply chain. According to the studies of Lowe (2020) due to the introduction of these barriers the time taken at the custom checks for the goods to pass has increased by 20% - 30% which has created a huge disruption for the products with **Just-in-Time** supply chain strategy as this model depends on receiving the products only when it's needed and the custom procedures has delayed this process which resulted in stock shortages and production delays. The studies of Fernandes et al. (2021) state that the supply chain cost has increased by 15% as the companies need to arrange more resources and Paperworks for the compliance with the

regulations and custom check procedures. According to the studies done by the Centre For European Reform (2021) the trade environment and volume will come back to the normal level like to that of Pre-BREXIT as companies are trying to adapt themselves with the new rules and regulations for the cross-border trade and establishing new strategies to make the supply chain more efficient and cost effective while aligning themselves with the latest regulations. Along with this, it will be really great for the long-term business if UK tries to extend their market in the other non-EU countries as that will make them less dependent on the EU and will be able to generate an excess income and will face less supply chain barriers compared to the trade barriers with EU.

4 Post-BREXIT Challenges in the Supply Chain -

Due to BREXIT the supply chain strategies of many companies have experienced huge disruptions which has increased the delays in delivering products and also cost of the supply chain has also increased due to major delays in acquiring the materials which has affected the production. The major problems that the companies are facing are custom delays, congestion at the border, sanitary and phytosanitary checks, compliance cost and supply chain strategies failures.

One of the most important and critical factors that has disrupted the supply chain is the custom check delays which are happening due to the new custom check procedures introduced after BREXIT. The findings of Lowe (2020) show that the companies are experiencing an increase of 20-30% more time compared to that of the Pre-BREXIT custom check timings which has increased the supply chain cost and the supply chain strategy which has been most affected from this change in the custom check procedures is the Just-in-Time (JIT) supply chain strategy which focuses on providing the products only when it is required but due to the delays in the new custom check procedures the JIT strategy is getting disrupted as a result the production delays and stock shortages are taking place in the companies who are using the JIT supply chain strategy.

Increase in the cost of the compliance has also increased in the post-BREXIT cross-border trade environment, according to the research of Fernandes et al. (2021) the trade sectors like automotive and agricultural has experienced an increase of about 15% operational cost as they had to spend more on employing the resources required to comply themselves with the latest regulations, tariff and non-tariff barriers and custom check procedures. The studies of Griggs and Shields (2020) show that the transportation cost of the companies has also increased due to the custom delays that are happening due to the new custom check procedures and the latest regulations like dual regulatory requirements. Lack of preparedness has also resulted in Port Congestion which caused delays in clearing the heavy loads of the products need for clearance, which increased the cost of operation and transportation and also loss of many goods might had happened as perishable goods were also a part of them and also caused delays in delivering products and stock shortages (Meyers et al. 2020).

4.1 Diversification of Supply Chain Post-BREXIT -

BREXIT has shown the companies of the UK the need of diversifying their supply chain as with the occurrence of BREXIT many industries which were highly dependent on the EU for the supply of materials has suffered a huge loss which made them rethink their supply chain strategy. To diversify their supply chain UK has signed many Free trade Agreements with non-EU countries like Australia, Canada and Japan which led to gaining various routes for free trade and decreases the dependency on the EU countries (Dept. For International Trade, 2021). The studies of Coyle et al. (2021) shows that the industrial sectors like automotive and pharmaceutical has extended their supply chain to these non-EU countries which will reduce their risk of getting affected due to the trade disruptions that are happening with the EU because of the BREXIT. UK's automotive sector was heavily dependent on the complex EU supply chain and the Just-In-Time supply chain strategy which resulted in the shortage of stock and huge losses due to the BREXIT. The studies of Lowe et al. (2020) shows that the automotive sector companies have started diversifying their supply chain

with a focus of supplying important parts from the domestic market and also from the global market apart from the EU market only which opened new routes of trade with the markets of different countries, one example of it is the Japan with whom UK firms have established a partnership which reduced their dependence on the EU market. The studies of Wheaton and Mills (2020) shows that the event of BREXIT has pointed out the vulnerabilities to the world of depending too much on a single market which resulted in proving that diversifying the supply chain places the businesses in better position and guarantees the smooth running of the operations.

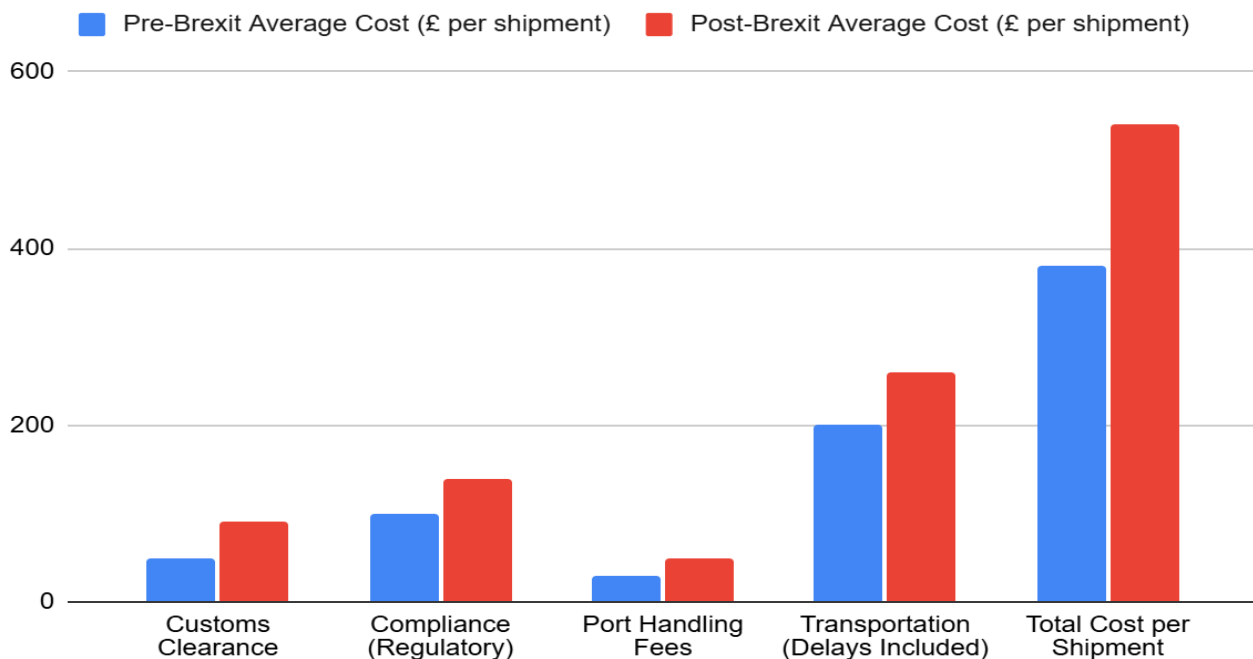
4.2 Supply Chain Adaptation based on the SCOR Model -

The SCOR Model (Supply Chain Operations Reference) is the model which is used to analyze, evaluate and improve the performance of the supply chain. It mainly consisted of five components that are – Plan, Source, Make, Deliver, and Return, all these components collectively represent the critical managing structure of a supply chain (APICS, 2020).

- Plan – Due to the Post-BREXIT trade environment companies have developed specialized demand forecasting techniques and supply chain plannings, according to the studies of Smith et al. (2020) the companies have depended heavily on the datasets to sharp their forecasting techniques and get an idea about the long-term trade environment due to the implications of the Post-BREXIT trade environment.
- Source – After BREXIT the companies are diversifying their supply chain by supplying materials from the non-EU countries as well which is reducing their dependency on the EU market minimizing the risks related to delays in deliveries and production and stock shortages, cross-border trades and aligning themselves with the latest regulations (Fernandes et al. 2021).
- Make – This component discusses about the process of assembling or making the products, in the post-BREXIT trade environment the companies had to readjust their supply chain and manufacturing points as due to the division of UK from EU many supply-chain disruptions took

place which resulted in production delays and stock shortages with diversification in the supply chain many new suppliers and markets opened up for the UK and companies had also opened up new manufacturing points in the EU region to gain access to the EU market without any barriers and have also developed global tie-ups which has increased the market radius. Zhang et al. (2021) in his studies has shown the changes in the strategies of the supply chain that have been done mainly in the automotive and electronics sectors.

- Deliver – It is one of the most crucial parts of the business mainly in retail and consumer goods sector, the studies of Harris & Smith (2020) show that in the post-BREXIT trade environment most of the companies are getting heavily dependent on the Third-Party Logistics and digital tracking systems to ensure on time delivery of the products and to navigate easily through the complex trade environment in the post-BREXIT scenario.
- Return – The process of returning products has become more complex in the post-BREXIT trade environment, according to the studies of Hughes & Jones (2020) the companies are adapting to the more streamlined strategies for the returns and the most concerning sectors are the fashion and e-commerce.



Supply Chain Cost Pre- and Post-BREXIT (ONS, 2021)

As seen in the above graph there has been a drastic increase in the cost of the supply chain in the post-BREXIT trade environment all the businesses have experienced a delays in the custom clearance, and also have found it hard to align themselves with the new regulatory policies for which they had to hire new resources which increased the operational costs of the companies. Due to lack of preparedness the ports of UK as also failed to handle the goods in the time which caused delays in the ports which resulted in expiration of many products as they were also carrying perishable goods and due to the complex supply chain strategies in the post-BREXIT trade environment the transportation and shipment cost has also increased. The studies of Thompson (2021) have shown that prior to the BREXIT the maximum delays that happened in the custom checks were 2 hours but in the post-BREXIT trade environment with the new custom check regulations some trucks were seen to wait for 8 hours at the Dover. The Pharmaceutical sector has also experienced a huge increase in the cost of the operations and regulatory cost as they had to go through dual regulatory requirements for publishing the medicines in both EU and UK (Koh et al., 2021).

4.3 Detailed Discussion of the challenges faced by Supply Chain in the post-BREXIT trade environment -

The analysis of the data of trade volume and the cost of supply chain in the pre- and post-BREXIT trade environment a significant degree of change can be noticed throughout all the industrial sectors of the UK including the agricultural, automotive and the pharmaceutical sector as well. The challenges that are causing this significant change in the cost of the supply chain are -

1. **Delays in Custom Checks** – In the post-BREXIT trade environment UK has witnessed delays in the custom check procedure due to the implications of newly formed regulations and the custom check policies, according to the data of Office of National Statistics UK has experienced a decrease of 40% in the rate of export just in the first quarter of the 2021 and the main reason for this was the new requirements for the custom check documents, cross-border trade rules and procedures and the supply chain strategy which got most affected by these was Just-in-Time strategy which focuses on supplying products when the demand arises, all these custom check delays has disrupted this strategy. According to the data from the Society of Motor Manufacturers and Traders the automotive companies had to stop their production due to the delays in supplying the required parts that was happening because of latest regulations and custom check procedures and documents required.
2. **Compliance Cost** – The pharmaceutical sector of the UK has suffered comparatively less to that of the other industrial sectors but the dual regulatory policy i.e. the medicines should be approved by both health sectors in the EU and UK to be published in the EU and UK market both which has increased the compliance cost and also the operational cost of the organizations as they had to hire new resources to align themselves with the latest regulations and the custom check policies and prepare the documents that are being mandated for the process. The agricultural sector has also experienced back lashes in terms of operational and compliance cost increase as new sanitary and

phytosanitary checks has been mandated for the products of this which made the whole cross-border checks more costly and also time consuming. The National Farmers Union has presented that there is an increase of 15% in the cost of compliance compared to the pre-BREXIT trade environment and the transportation cost of the goods has also increased due to delays in the custom check procedures and all together which used to be 50 GBP per shipment cost in the pre-BREXIT trade environment to 90 GBP per shipment in the post-BREXIT trade environment.

3. Overcrowding/Congestion at the Ports – Port congestion has always been a major visible issue from the beginning of the post-BREXIT trade environment. The data of the UK Road Haulage Association has shown that the trucks had to wait for a long time for custom clearance because of the newly developed custom check procedures and the policies and because of lack of preparedness as well. These delays and procedures have also increased the transport cost, rerouting and the delays by 30%, which used to 200 GBP per shipment in the pre-BREXIT trade environment has increased to 260 GBP per shipment in the post-BREXIT trade environment.

4.4 Importance of the Third-Party Logistics Provider (3PL) -

In the complex post-BREXIT trade environment, the third-party logistics providers are playing a huge role in helping the companies navigate through the complex trade policies, cross-border trade and the custom checks, these third-party logistics providers offer a range of services from preparing documents for smooth custom check procedures to digital tracking of the products to check in which stage they are at in the custom clearance procedure. The two companies which have streamlined the supply chain and custom check procedures for most of the companies in UK are DHL and Kuehne + Nagel these companies have provided their clients exemplary

custom check procedures and also have reduced the time taken in generating leads in the cross-border trade.

An example of the services by the DHL and the Kuehne + Nagel is the service package which they have specially made for the cross-border trade in the post-BREXIT trade environment, in this package they have streamlined the supply chain for the companies which has reduced the time taken at the custom checks and also provides real time tracking of the shipment which helps the companies to take decisions based on the situations like potential delays or efficiency of the supply chain.

5 Latest Strategies that the major Import and Exports sectors are Implementing -

BREXIT has forced the Businesses to rethink their supply chain strategies and the operational system of the product deliveries and the way of navigating the trade environment. In the complex post-BREXIT trade environment, many new regulations were imposed, and new custom check procedures and documentation were made mandatory which increased the operational cost of the companies and the transportation cost as well as due to lack of preparedness port congestions were taking place for which trucks with shipments had to wait for a longer time which increased the transportation costs, among the goods many perishable goods were also present from the agricultural sector which got affected due to the delays in the custom checks for which companies in the agricultural sector also had to bear a certain amount of losses. Along with the agricultural sector, the automotive sector and the financial sector have also suffered huge losses as before BREXIT they used to enjoy barrier free trade with easy access to the EU market but with the introduction of BREXIT many barriers have formed in the cross-border trade and these barriers have made the businesses form new strategies for the supply chain so that the businesses can perform their operations without any disruptions. The new strategies based on the different sectors is discussed in the following sections,

5.1 Strategies in the Automotive Sector -

The first strategy that the companies in the Automotive sector have taken is localizing the supply chain. Companies in the automotive sector used Just-In-Time supply chain which has been disrupted in the post-BREXIT trade environment due to the custom delays and the tariff and non-tariff barriers. In the new custom check regulations, rules of origin rule have also been involved which has caused significant disruptions in supplying the parts from the EU, Localizing the supply chain will reduce the dependency of the UK automotive sector on the EU market for the supply of the parts for manufacturing. The strategy consists of sourcing materials from the domestic market and from the non-EU countries which will help the companies get rid of the custom delays and the tariff and non-tariff barriers, but it will also increase the overall cost of the production as compared to the EU suppliers the price of the parts in the domestic and global market is higher which increases the overall production and operational costs but compared to the delays and stock shortages which increased the cost more comparing to the total average cost of the new supply chain.

The second strategy which the companies are taking to get access to the EU market is that they are making EU based operational facilities which will help them to access the EU suppliers for supplying the parts for manufacturing easily. Establishing the manufacturing points in the EU will help the companies access the market of EU and perform trade without any tariff, non-tariff and custom check barriers (Davies & Studnicka, 2018). For example, Nissan has created facilities at Spain and France to increase their presence in the EU market which helped them to tackle the cross-border trade delays and custom check procedures and costs related to it (Davies & Studnicka, 2018). The third strategy that the companies in the UK and the UK government are taking is diversifying the market i.e. they are signing free trade contracts with many non-EU countries like Australia, Japan, and Canada, one of the biggest examples of it is the UK-Japan Comprehensive Economic Partnership Agreement which will provide free trade opportunities and reduced tariffs on the export of UK cars to Japan which will provide opportunities for developing

global trade and will reduced the dependency of UK automotive sector on the EU market.

5.2 Strategies implemented by the companies in the Agricultural Sector -

The first strategy that the companies in the Agricultural sector of UK are taking is Diversifying the Market, the studies of Sampson (2019) show that the firms had been hit had with the new regulatory policies like sanitary and phytosanitary checks as this has led to increase of the compliance cost which many farmers cannot afford in the UK market and for this reason they are searching market to export their products and diversify their supply chain as it will reduce their dependency on the EU market alone for exporting the goods. The research of Sampson (2019) shows that the agricultural sector has started exporting their products to the United States, Canada and China which has opened opportunities of global trade for them, but they come with their own set of challenges like meeting the local regulatory requirements. The second strategy that the Agricultural sector has taken is Investing in the compliance and logistic infrastructure which will help the farms in the UK to meet the regulatory requirements in both the UK and EU and this will help in easily performing the sanitary and phytosanitary checks which will reduce the custom delay in the cross-border trade. They have also installed refrigerators which will help them to store the perishable goods, which will help in decreasing the cost of the losses due to the expiration of the perishable goods. Along with these some farms have also started using vertical integration strategies which has converted the whole processes of producing and packaging under a single roof which has made it easy for the farms to perform sanitary checks and control the supply chain as and when required without any delays making the cross-border trade less time consuming. The third Strategy that the Agricultural Sector has implemented is signing/forming trade partnerships with the non-EU countries which will help them to expand their export market in the global market beyond EU and will reduce their dependency on the EU market for the export trade. The Australia-Free trade agreement has also provided opportunity for exporting a range of

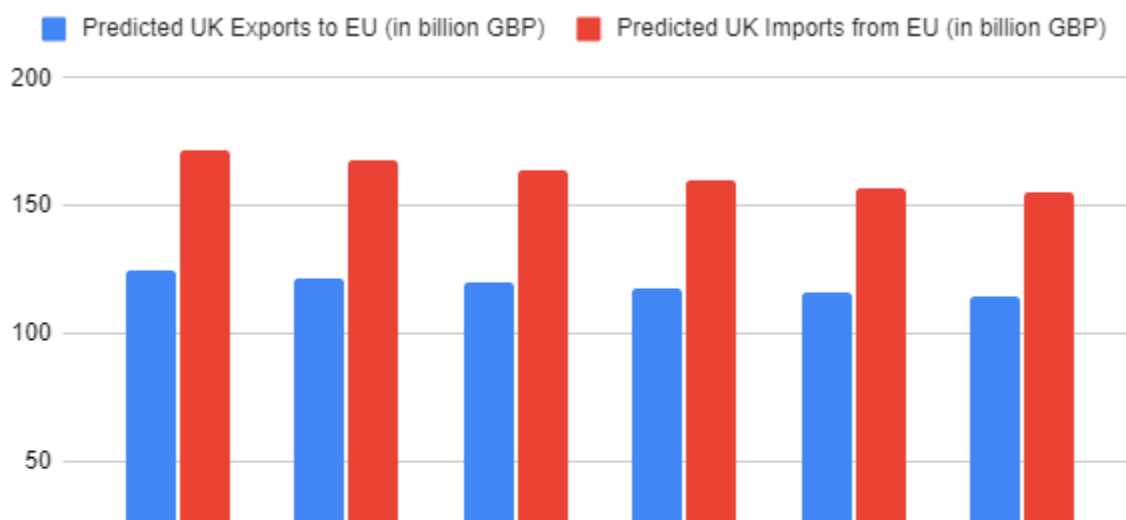
agricultural products without any tariff which provided British Farmers new market for trade and reduced dependence on the EU market.

5.3 Strategies implemented by the Financial Services Sector Companies in UK -

The first strategy that the companies in the Financial Services sector of UK have taken is the Relocation of their Operational unit in the EU countries, according to the data from the Centre for European Reform (2023) with the loss of the passporting right the financial services companies has lost their right of work in the EU market in a hybrid manner i.e. without establishing their branches in EU they cannot extend their services in the EU regio for this reason many financial services companies established their operational offices at the EU countries to get the huge market of the EU, more than 300 financial services companies have established some parts of their operational offices and duties in the EU region to get access to the EU market and the clients. The second strategy that the companies have used is the Diversification of the Markets i.e. the companies has started extending their services to the non-EU countries as well along with the EU countries which will reduce the dependency of the companies on the EU market. The markets in which the companies have extended their services are United States, Japan and Singapore, they are also trying to extent their markets in the Southeast Asia region as well. The third strategy that the financial services companies started using was digitalization of the banking processes and application of Fintech in the Financial Services which helped the companies expand their reach in the areas where they cannot reach physically but with the use of the digitalization and fintech they can serve the clients remotely and usage of the blockchain has also made this whole process more accessible and secure.

6 Long-Term Effect of BREXIT on the Cross-Border trade between UK and EU -

As UK is in the beginning phase of the BREXIT so making a conclusion about its long-term effects on the UK is not ideal at this point, though some economists argue that the loss of the EU will have a long lasting impact in the economy of the UK as it played a huge role in the contribution of the GDP of UK, while some economists state that UK will recover in the long run with their trade partnership with the major countries with huge market outside the EU region and some examples of it are United States, Japan, and Canada. The research of London School of Economics shows that with the loss of the EU market due to BREXIT the UK has experienced a loss of 4% in the GDP compared to that of the pre-BREXIT GDP of UK. The studies of Crafts (2018) shows that the new trade agreements with the non-EU nations will not only help the UK to coverup the losses but will also provide exposure to the global market and might earn a huge profit compared to that of the sales in the EU market. The reports of the International Monetary Funds have also showed a decrease in the inflow of the Foreign Direct Investment as before BREXIT UK used to be the first choice for the companies to establish operational centers to access the huge market of the EU without any barriers or regulatory restrictions but after BREXIT many new regulations are imposed and tariff and non-tariff barriers, custom checks have made it difficult for the businesses to access the EU market easily due to this major companies has stopped their investment in developing the operational branches in the UK due to which UK has experienced a decrease in Foreign Direct Investment. According to the data from the Office for Budget Responsibility, UK trade policy Observatory and Organization for Economic Co-operation and Development, they predicted that there will be a 18% decrease in the export and import trade volume in the coming years till 2030 and will be slow increase in the GDP of both EU and UK.



7 Conclusion and Recommendations -

7.1 Key Findings -

In this dissertation many critical factors have been examined about the Impact of BREXIT on the Export and Import trade between EU and UK with main focus on the change in trade volumes in the pre- and post-BREXIT trade environment, changes/implementation of new regulations, supply chain strategies implemented by the most affected business sectors and the long-term effect of the BREXIT on the UK and EU trade relation. When it comes to the trade volumes in chapter 3 a detailed analysis of the trade volume before and after BREXIT had been performed which showed that in the first quarter of 2021 itself 40% decrease was noticed in the export and import trade volume with the EU. In chapter 4 detailed analysis of the supply chain challenges faced by the most affected sectors of the businesses had been covered and also the importance of the Third-Party Logistic provider had also been discussed in detail where it showed that the main factors that disrupted the trade flow and the supply chain in the post-BREXIT trade environment were the latest custom check

procedure and the regulations and documentation required, other barriers include tariff and non-tariff barriers and all these things resulted in port congestions and custom check delays the research of Thompson (2021) shows that there is a significant increase in the custom delays in the post-BREXIT trade environment compared to that of pre-BREXIT trade environment. In chapter 5 detailed discussion of the sector wise supply chain strategies that the companies have taken to tackle with the complex post-BREXIT trade environment had been done where one of the most common strategies that companies from all the sectors have used is to expand their market to the non-EU countries as well to reduce their dependency on the EU alone and also to reach the global market. In chapter 6 a detailed discussion of the long-term effect of the BREXIT on the UK had been done, though it is not possible to predict the long-term effect of BREXIT as it is in the budding stage but according to some economists there will always be an impact on the UK GDP for leaving the easy access of the EU market by separating itself from EU while some other economists predict that by entering the global market UK will perform better compare to the only access of the EU market and will perform exponentially better in the long run.

7.2 Recommendations for the Policy Makers and the Businesses -

Through the critical findings from this dissertation many recommendations can be made to the Businesses and the policy makers but some of the most important recommendations for the businesses and policy makers of all the sectors are

- Investment in the Compliance and the Logistics of the Companies as it is very important for the UK to understand the custom check procedures and documents and the tariff and non-tariff if they want to maintain their hold in the EU market.
- Diversification of the market as it will reduce their dependency on a single market and losing some share of the market will not heavily affect the business.

- Diversification of the Supply chain or localization of the supply chain as this will increase the business in the domestic market and will also cause less to null disruptions due to cross-border trade delays, though it can be costly compared to the imports from the EU but the amount of delay and the disruption is happening and the loss that the companies are taking for that is considered to be more than the cost of the localization of the supply chain.

7.3 Scope for Future Research -

Occurrence of BREXIT is still a new event that have left both the UK and the EU in a complex trade relation and the EU is developing many trade regulations and also supporting UK to maintain a stable trade relation with the EU and gain the access to the huge EU market but the companies in the UK are facing disruptions in the supply chain due to custom delays and the tariff, non-tariff barriers which has led to decrease in the export and import trade volume between the EU and UK, and all these issues provide various topics to perform future research on this piece of work and topic but the most suitable or beneficial topics according to my research and preference will be the Long-Term effect of BREXIT on the UK businesses and also the SME's adaptation to the complex post-BREXIT Trade environment as they face many different challenges along with the ones that large companies are facing.

Conclusion -

Overall to sum things up, BREXIT has brought significant challenges for the businesses in the UK but there is still a scope of adaptation to these challenges by investing in the factors that are responsible for the growth of the businesses in the UK like trade partnerships, compliance and logistics, expanding the reach of the businesses to the global market and so. The strategies implemented by the UK government and also by the businesses in the UK will determine the path of the success for the import and export trade of the Businesses in the UK, their ability to navigate the complex post-BREXIT trade environment will define the route of success for them.

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Appendix –

Ethics Form

Module Code - BUSI-1359-M08-2023-24

Project Title - Investigating The Impact Of BREXIT On The Export And Import Trade Between EU And UK

Brief Description -

Research Aim/ Objective: - This research aims to provide a thorough analysis of the impact that Brexit has left on the import and export trade between United Kingdom (UK) and European Union (EU). This study aims to provide a knowledge of the short-term and long-term effects of Brexit on the trade volumes, the strategies of the companies are putting into effect to tackle Brexit and also the new trade policies, rules and regulations that are being put in place.

Research Methods: - I will be using both the quantitative and qualitative approaches to collect secondary data and perform a thorough analysis for a proper understanding of the topic. Mainly secondary data collection and analysis will be performed, and all data will be collected by keeping the “Ethical Regulation” as our top priority and data will only be collected from credible sources like Journals, Official UK Government websites and also Office for National Statistics and World Bank. I will also try to gather and analyze the

interview videos of the higher officials who have given any interviews related to Brexit and the new rules and regulations which are being imposed.

Start Date of Data Collection – 12 September 2024.

End Date of Data Collection – 31 October 2024.

Data Collection Method – Secondary Data Collection.

For data collection involving human participants:

I will use the required Participant Information and Consent templates:

Use authorised PIF
and Consent Required
field
☐ Yes
☒ NA

I will not collect data from university staff, children or vulnerable people:

No vulnerable
participants Required
field
☐ Yes
☒ NA

I will collect only the minimum personal data necessary to answer my research questions:

Minimum personal
data Required field
☐ Yes
☒ NA

Any in-person contact with participants will only occur in quiet public spaces:

Quiet public places
Required field

- ☐ Yes
☒ NA

I will not collect the identities of people who respond to questionnaires (anonymity):

Anonymity Required
field

- ☐ Yes
☒ NA

I will not include or report the names of individuals alongside data I store about them (confidentiality):

Confidentiality
Required field

- ☐ Yes
☒ NA

For all data collected:

I will store data only on my University G: drive or University OneDrive and not on mobile devices, memory sticks or laptops:

Data protection
Required field

- ☒ Yes
☐ NA

I will delete all collected data at the end of this academic year:

Delete data Required
field

☒ Yes

☐ NA